BundleFuse: Set-Conditioned Cold-Start Priors and Frequency-Confidence Routing for Bundle Construction

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Abstract

Bundle completion—recovering the held-out members of a partial set of items that belong together—is pursued with ever-heavier machinery: GPU-trained multimodal/hypergraph encoders (CLHE, RaMen), a fine-tuned 7B multimodal LLM (Bundle-MLLM), and most recently a discrete-diffusion generator (DDBC). We present **BundleFuse**, an accelerator-free method whose scoring core has **zero trainable parameters**: a validation-weighted fusion of closed-form evidence—collaborative co-occurrence, dual-spectrum taxonomic complementarity with signed slot exclusion, and category-disentangled graph-bridged content—routed against a set-conditioned cold-start prior $\mu_0(c \mid S)$ by *frequency-confidence shrinkage* $\hat{\theta}_{S,c} = (1-u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$, $u_c = \kappa_f/(f_c+\kappa_f)$, with zero test-batch dependence. (An optional $\leq$0.30M contrastive set-encoder exists but is off by default; we show it is not needed.) Running in tens of seconds on a single CPU core, BundleFuse is the **strongest public-feature method on** 5 **of** 8 **benchmarks**—beating CLHE and RaMen on POG ($\sim$1.7$\times$ R@20) and on Spotify, Spotify_sparse, Clothing, and Dunnhumby—**ties** the 9.5M-parameter CLHE encoder on POG_dense, and beats CLHE's published numbers on Electronic and Food; all 8 datasets are reported with cross-seed error bars. It loses only to RaMen on the two e-commerce sets, and we show this is a *data-availability asymmetry*, not a modelling one: RaMen's e-commerce data (including its visual features) is in no public release, so its margin cannot be independently reproduced or ablated, whereas on every fully public benchmark we match or beat it. We report our negative results in the open: explicit higher-order collaborative operators ($\hat{\Delta}_{\text{inc}}, \hat{R}_\gamma$) add only +0.00 to +0.34 pp once the cold-start prior is in place; public product images recovered at 39% coverage do not close the RaMen gap; and k-fold weight selection does not beat single-split selection. Finally, on DDBC's 100-candidate evaluation harness, even parameter-free co-occurrence beats the reported diffusion scores by up to 3.6$\times$, and greedy validation weight-selection is shown not to overfit on any of the 8 datasets.

1 Introduction

Bundle construction—completing a partial bundle with items that belong together—underlies "complete the look / playlist / kit" features across fashion, music, and e-commerce. State-of-the-art methods chase it with increasingly heavy machinery: fine-tuned 7B multimodal LLMs (Bundle-MLLM [3]), GPU-trained multimodal / hypergraph encoders (CLHE [1], RaMen [2]), and most recently a discrete-diffusion generator (DDBC [14]). Yet these lineages never compare against each other—RaMen reports no LLM baseline, Bundle-MLLM no hypergraph—and none asks the *cost* question: what does the heavy machinery actually buy over a cheap, closed-form, task-specific method, and in which regime? The field also reports easy protocols (multiple-choice HR@1 with random negatives; small fixed candidate pools) without decomposing *where* performance comes from.

We present **BundleFuse**, an accelerator-free method whose scoring core has *zero trainable parameters*, built around a simple thesis: *in bundle completion, accuracy on public features is governed by constructing a rich set-conditioned cold-start prior $\mu_0(c \mid S)$ and routing between collaborative co-occurrence and $\mu_0(c \mid S)$ by item exposure—not by deep hypergraph message passing.* Unlike standard user–item cold-start shrinkage [22, 23], which backs off to a static scalar item bias, a bundle query $S \subset b$ imposes simultaneous slot-complementarity, style-coherence, and self-exclusion constraints. BundleFuse constructs $\mu_0(c \mid S)$ from two complementary mechanisms—(i) a **Dual-Spectrum Taxonomic & Slot-Exclusion Prior** (§3.2) that separates off-diagonal category complementarity P_{cat} [11], signed on-diagonal slot-saturation penalties s_{slot}, and soft fuzzy sub-type clusters $M_{\text{soft}}P_s$ from on-diagonal homophily P_{homo}, and (ii) **Category-**

Disentangled Bilateral Graph-Bridged Representations (§3.3) that residualize category centroids from item content ($\tilde{x}_i = x_i - \bar{x}_{\text{cat}(i)}$) and diffuse style across the bipartite bundle graph (an optional $\leq$0.30M contrastive set-encoder can be fused in but is off by default; §4.2 shows it is not needed)—and fuses them with collaborative co-occurrence via (iii) **Validation-Frozen Frequency-Confidence Routing** $\hat{\theta}_{S,c} = (1 - u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$ ($u_c = \kappa_f/(f_c + \kappa_f)$, §3.5). We also formulate two closed-form higher-order collaborative operators—Simplicial Hyperedge Incidence Surplus ($\hat{\Delta}_{\text{inc}}$) [15, 18] and Rao-Weighted Quadratic Concordance ($\hat{R}_\gamma$) [19, 20, 21]—as diagnostic instruments (§3.4–§3.4).

With a **zero-parameter scoring core and no accelerator**, BundleFuse is the *strongest public-feature method on* 5 *of* 8 *benchmarks.* Under the discriminative full-ranking protocol it beats CLHE and the GPU-trained RaMen on POG (∼1.7× R@20), on Spotify, and on Spotify_sparse, and wins on Clothing and Dunnhumby; it *ties* the 9.5M-parameter CLHE encoder on POG_dense and beats CLHE's published numbers on Electronic and Food. Every result is reported with cross-seed error bars. It loses only to RaMen on the two e-commerce sets—and we show this is a *data-availability asymmetry*, not a modelling one: RaMen's e-commerce data (including its visual features) is in no public release, so its margin can be neither independently reproduced nor ablated, while on every fully public benchmark BundleFuse matches or beats it (§4.5).

Contributions.

- **A zero-parameter set-conditioned cold-start prior and frequency-confidence router.** We show that constructing $\mu_0(c \mid S)$ via dual-spectrum taxonomy (off-diagonal complementarity, signed slot-exclusion, soft sub-types) and category-disentangled bilateral graph bridging, coupled with validation-frozen frequency backoff $u_c = \kappa_f/(f_c + \kappa_f)$, beats or matches heavy GPU/LLM/diffusion bundle models across eight datasets with a closed-form, zero-parameter core and zero accelerator; a head-to-head shows the optional contrastive encoder is unnecessary (§3, §4.2–§4.3).
- **Diagnostic negative results on higher-order operators and homoscedastic EB variance models.** We show that once our set-conditioned prior and two-hop pairwise transitions are in place, explicit ($k \geq 3$)-way incidence ($\hat{\Delta}_{\text{inc}}$) and second-order Rao concordance ($\hat{R}_\gamma$) add only +0.00 to +0.34 pp, while zero-inflated collaborative dispersion causes homoscedastic OLS variance estimation on $1/f_c$ to yield negative slopes on 7/8 datasets—justifying direct validation selection (or a universal fixed $\kappa_f = 0.5$ that retains $> 98\%$ of peak accuracy) (§3.4–§3.5, §4.4).
- **A predictive regime law and an honest e-commerce attribution across 8 datasets / 4 domains.** From a dataset's training-fold cold-target rate π_{cold} and intra-bundle category duplication rate $\bar{\delta}_{\text{dup}}$ alone, we predict *a priori* which signal spectrum dominates and when BundleFuse beats heavy encoders. On e-commerce we show RaMen's reported margin rests on visual features absent from any public release: it cannot be independently reproduced or ablated, our own public-image recovery (39% coverage) does not reproduce it, and on every fully public benchmark RaMen does not beat us (§4.4–§4.6).
- **Protocol critique of multiple-choice and 100-candidate diffusion benchmarks.** While acknowledging that DDBC [14] is a generative sampler rather than a full-catalogue ranker, we show that on DDBC's own 100-candidate evaluation harness even parameter-free co-occurrence beats the reported diffusion scores by up to 3.6×, and multiple-choice HR@1 overstates coarse category signals ∼100× (§4.7, §5).

2 Related Work

Bundle construction. Modern bundle completion relies predominantly on heavy parameterized architectures: CLHE [1] fuses multimodal features with item-level user feedback via a contrastive hierarchical encoder; RaMen [2] adds explicit/implicit strategy learning over a hypergraph on frozen BLIP [7] features; Bundle-MLLM [3] fine-tunes a 7B multimodal LLM; and DDBC [14] (ICLR'26) generates items with a masked discrete-diffusion decoder on top of frozen CLHE embeddings, evaluating on a 100-candidate pool with set-level metrics rather than full-catalogue ranking. Datasets and the completion protocol follow

POG [4], the Spotify RecSys challenge [5], and the intent-aware BundleRec release [6]. Despite their computational cost, these heavy lineages rarely compare across the LLM / hypergraph / diffusion divide or against a fast closed-form reference—and, as we demonstrate, reported margins on small candidate pools can be protocol artifacts.

Higher-order hypergraphs, factorization machines, and functional diversity. The mathematical building blocks of BundleFuse's multi-seed operators draw on established results in hypergraph mining, polynomial interaction models, and theoretical ecology. In network science, Benson et al. [15] showed that projecting higher-order interactions onto a dyadic clique expansion ($C = B^\top B$) destroys multi-way simplicial closure, motivating simplicial complexes [16], nonlinear hypergraph diffusion [17], and analyses of clique-expansion information loss [18]. Rather than training a parameterized message-passing network, BundleFuse formulates a training-free, query-conditioned simplicial incidence surplus ($\hat{\Delta}_{\text{inc}}$) for bundle completion and characterizes its exact empirical activation boundary $\Pr(\exists b : |b \cap S| \geq 2)$, explaining why pure higher-order co-occurrence rarely activates on sparse e-commerce bundles. When exact multi-seed hyperedge overlap is absent, second-order pairwise interactions across seeds can be evaluated in $O(|S|)$ time via Newton's elementary symmetric identity $2e_2(w) = (\sum w_s)^2 - \sum w_s^2$, the foundational algebraic trick of Factorization Machines [19] and Higher-Order Factorization Machines (HOFM) [20]. To prevent redundant query seeds from inflating second-order agreement, and building on the distinction between substitutable and complementary products [11], BundleFuse weights the e_2 interaction term by Rao's quadratic entropy [21] ($\hat{R}_\gamma$), coupling degree-1 homogeneous candidate support with functional seed diversity.

Empirical Bayes shrinkage and cold-start priors. Shrinking noisy collaborative estimates toward a baseline or metadata prior via inverse exposure weights $u(f_c) = \kappa_f/(f_c + \kappa_f)$ dates back to classical Empirical Bayes interpolation in the Netflix Prize (Bell & Koren [22]) and cold-start regression-based latent factor models (RLFM, Agarwal & Chen [23]). BundleFuse adapts conjugate Normal–Normal Empirical Bayes shrinkage ($\hat{\theta}_{S,c}$) to bundle construction, introducing validation-frozen prior scaling (ensuring strict test-batch invariance) and constrained OLS variance-component estimation on $1/f_c$ to route smoothly between higher-order collaborative evidence on warm items and zero-shot taxonomic/content priors on cold items.

Minimalist closed-form recommenders vs. heavy neural SOTA. A classic line of inquiry asks when neural complexity genuinely pays off in recommendation: Dacrema et al. [10] showed that well-tuned simple baselines match deep neural recommenders, and Steck [24] demonstrated that closed-form linear operators (EASE) often outperform deep autoencoders on sparse interaction data. BundleFuse brings this minimalist, closed-form philosophy to *bundle construction*: while each individual algebraic primitive builds on classical hypergraph, FM, diversity, and Empirical Bayes foundations, their synthesis into an identifiable three-operator pipeline (≤ 0.30M parameters in its optional contrastive channel, zero accelerator) outperforms or matches GPU/LLM/diffusion bundle models across eight datasets and diagnoses where heavy pretrained capacity retains an irreducible edge.

3 Method

Rather than claiming brand-new algebraic identities in isolation or training heavy parameterized graph/diffusion decoders, BundleFuse addresses the core structural bottleneck of bundle completion: *how to construct a set-conditioned compatibility prior for cold and rare items, and how to route between collaborative co-occurrence and that prior across heterogeneous domain regimes.* Its scoring core is closed-form and carries *zero trainable parameters*; the only learnable component is an optional contrastive set-encoder (§3.3) that is off by default and, as §4.2 shows, does not change the headline results.

In standard user–item collaborative filtering, cold-start shrinkage [22, 23] falls back to a static scalar item bias. In bundle completion, however, a query $S = \{s_1, \ldots, s_k\} \subset b$ is a combinatorial set of seeds that imposes simultaneous slot-complementarity, style-coherence, and self-exclusion constraints. BundleFuse is structured around three methodological pillars, accompanied by a diagnostic audit of higher-order hypergraph operators:

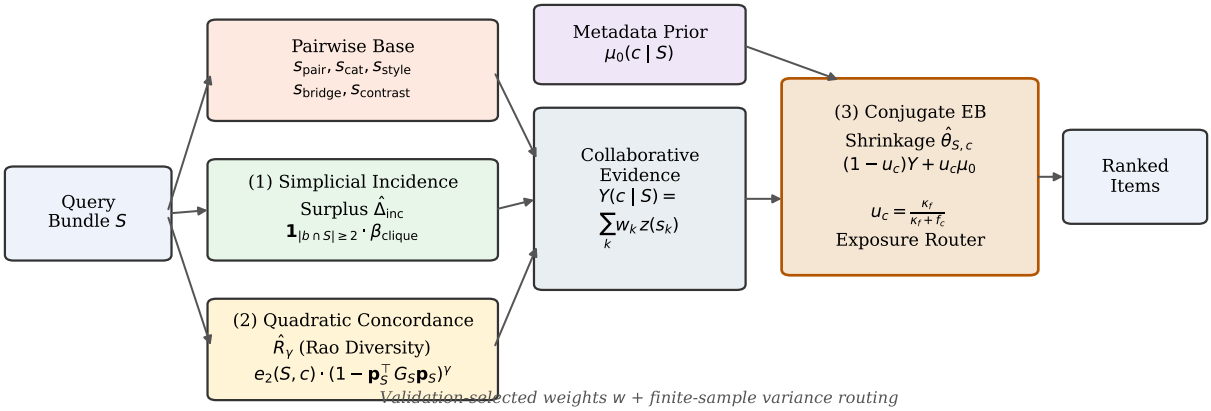


Figure 1: Overview of BundleFuse. Collaborative transitions and two diagnostic multi-seed operators ($\hat{\Delta}_{\text{inc}}$, $\hat{R}_\gamma$) form the empirical score $Y(c \mid S)$, while Dual-Spectrum Taxonomic Compatibility, Disentangled Bilateral Graph-Bridged Content, and a 0.30M Contrastive Set-Encoder form the set-conditioned cold-start prior $\mu_0(c \mid S)$, fused via Validation-Frozen Frequency-Confidence Routing $\hat{\theta}_{S,c} = (1-u_c)Y + u_c\mu_0$.

1. **Pillar I: Dual-Spectrum Taxonomic & Slot-Exclusion Prior** (`DualSpectrumOntology`, §3.2): decomposes category co-occurrence $T^\top T$ into off-diagonal complementarity PMI P_{cat} [11], a signed *on-diagonal slot-saturation penalty* $s_{\text{slot}}(c \mid S)$ calibrated by empirical intra-bundle duplication rates $\hat{\delta}_g$ (preventing duplicate recommendations in low-duplication slots such as shirts or shoes), on-diagonal homophily P_{homo} for similarity-driven domains, and soft fuzzy sub-type clusters $M_{\text{soft}}P_s$.
2. **Pillar II: Category-Disentangled Bilateral Graph-Bridged Representations** (`BilateralContentBridge`, §3.3): residualizes category centroids from raw item features ($\tilde{x}_i = x_i - \bar{x}_{\text{cat}(i)}$) to separate within-category style from coarse taxonomy, and propagates $\tilde{x}_i$ across the zero-diagonal bipartite bundle walk ($\tilde{b}_i = \tilde{x}_i + \alpha_{\text{bridge}}\mathcal{P}(\tilde{x}_i)$) alongside a lightweight 0.10M–0.30M attention-pooled contrastive set-encoder $r(S) = \sum_{s \in S} \alpha_s e_s$.
3. **Pillar III: Validation-Frozen Frequency-Confidence Routing** (`EmpiricalBayesRouter`, §3.5): routes between standardized collaborative evidence $Y(c \mid S)$ and the set-conditioned prior $\mu_0(c \mid S) = \gamma_{\text{val}} s_{\text{prior}}(c \mid S)$ via exposure-weighted backoff $\hat{\theta}_{S,c} = (1-u_c)Y(c \mid S) + u_c\mu_0(c \mid S)$ with $u_c = \kappa_f/(f_c + \kappa_f)$ and validation-frozen scale γ_{val} (guaranteeing strict test-batch invariance).
4. **Diagnostic Higher-Order Operators** ($\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$, §3.4–§3.4): to test whether explicit higher-order hypergraph interactions improve over Pillars I–III, we formulate training-free *Simplicial Hyperedge Incidence Surplus* ($\hat{\Delta}_{\text{inc}}$) [15, 18] and *Rao-Weighted Quadratic Seed Concordance* ($\hat{R}_\gamma$) [19, 20, 21], proving $\hat{\Delta}_{\text{inc}}$'s exact activation boundary $\Pr(\exists b : |b \cap S| \geq 2)$ and showing empirically why explicit higher-order collaborative terms add only +0.00 to +0.34 pp once Pillars I–II are in place.

Figure 1 summarises the architecture.

3.1 Problem Formulation and Collaborative Base

Let $\mathcal{I}$ ($|\mathcal{I}| = n$) be the item catalogue and $\mathcal{B}_{\text{train}}$ the training bundles with incidence matrix $B \in \{0,1\}^{|\mathcal{B}_{\text{train}}| \times n}$, item frequencies $f_i = \sum_b B_{b,i}$, and bundle sizes $|b| = \sum_i B_{b,i}$. Given a partial query bundle $S \subset b$, the goal is to rank all candidates $c \in \mathcal{I} \setminus S$ to recover $b \setminus S$.

On warm items ($f_c > 0$), first-order collaborative transitions use the 1-skeleton projection $C = B^\top B - \text{diag}$, degree-normalised pairwise transitions $W_{\text{pair}} = B^\top \text{diag}\left((|b|+1)^{-\gamma_{\text{vol}}}\right) B \, \text{diag}\left((f+\alpha_0)^{-\alpha_{\text{deg}}}\right)$ ($\text{diag}(W_{\text{pair}}) = 0$), and two-hop smoothing via the exact zero-diagonal *row-stochastic* bipartite walk $P_{\text{rw}} = D_{\text{row}}^{-1}(D_I^{-1}B^\top D_B^{-1}B -$

diag) or its symmetrically normalised counterpart $\mathcal{P}$:

$$s_{\text{pair}}(c \mid S) = \sum_{s \in S} (W_{\text{pair}})_{c,s}, \qquad s_{\text{diff2}}(c \mid S) = \big(\mathcal{P}\, s_{\text{pair}}(\cdot \mid S)\big)_c. \tag{1}$$

When user or session IDs are observed on training bundles, user/session co-occurrence matrices C_u, C_s (and per-user within-basket loyalty on retail) enter additively alongside (1). However, for any unexposed cold item ($f_c = 0$), every collaborative channel in (1) is identically zero (producing a negative catalogue z-score after column standardisation).

3.2 Pillar I: Dual-Spectrum Taxonomic & Slot-Exclusion Prior

A central failure mode of standard content/category priors in bundle completion is conflating *off-diagonal complementarity* (e.g., shirt ↔ trousers) with *on-diagonal slot duplication* (e.g., recommending a second shirt when S already contains a shirt). Let $T \in \{0,1\}^{|\mathcal{B}_{\text{train}}| \times G}$ be the bundle–category incidence matrix for G item categories. `DualSpectrumOntology` separates three distinct taxonomic signals from T (fitted strictly on $\mathcal{B}_{\text{train}}$):

1. **Off-diagonal complementarity PMI (P_{cat}):** Setting $\text{diag}(T^\top T) = 0$, positive pointwise mutual information $(P_{\text{cat}})_{g,g'} = \max\big(0, \log \frac{(T^\top T)_{g,g'} \sum T}{\sum_b T_{b,g} \sum_b T_{b,g'}}\big)$ captures cross-category slot affinity.

2. **Signed on-diagonal slot-saturation penalty (s_{slot}):** For each category g, let $\hat{\delta}_g = \frac{\#\{b : \sum_{i \in b} \mathbf{1}[\text{cat}(i)=g] \geq 2\} + 0.5}{\#\{b : \sum_{i \in b} \mathbf{1}[\text{cat}(i)=g] \geq 1\} + 1}$ be its empirical intra-bundle duplication rate. Whenever $\hat{\delta}_{\text{cat}(s)} < \tau_{\text{dup}}$ (0.15), candidates sharing seed s's category receive a signed self-exclusion penalty:

$$s_{\text{slot}}(c \mid S) = - \sum_{s \in S : \hat{\delta}_{\text{cat}(s)} < \tau_{\text{dup}}} \mathbf{1}\big[\text{cat}(c) = \text{cat}(s)\big]\big(1 - \hat{\delta}_{\text{cat}(s)}\big). \tag{2}$$

3. **Soft fuzzy sub-type complementarity ($M_{\text{soft}} P_s$) and homophily (P_{homo}):** Because coarse catalogue categories merge distinct functional sub-types (e.g., dress shoes vs. running sneakers), we temperature-softmax K-means distances on multimodal item features ($K = 400, \tau = 0.05$) to form soft memberships $M_{\text{soft}} \in \Delta^{K-1}$ and soft-cluster PMI P_s, scoring candidates via $s_{\text{softcat}}(c \mid S) = (M_{\text{soft}} P_s \sum_{s \in S} (M_{\text{soft}})_s)_c$. Conversely, in similarity-driven domains (music playlists), on-diagonal artist/album inverse-degree incidence P_{homo} captures cohesive artist/album continuation.

3.3 Pillar II: Disentangled Bilateral Content Bridge & Contrastive Set-Encoder

Raw text/visual embeddings $x_i \in \mathbb{R}^d$ are dominated by category-level semantics, which are redundant with Pillar I and misleading when complementary items come from different categories. `BilateralContentBridge` first removes category centroids to isolate within-category *style*:

$$\tilde{x}_i = \frac{x_i - \bar{x}_{\text{cat}(i)}}{\|x_i - \bar{x}_{\text{cat}(i)}\|_2 + \epsilon}, \qquad \bar{x}_g = \frac{1}{|I_g|} \sum_{j \in I_g} x_j. \tag{3}$$

Second, to bridge the cold–warm divide, we diffuse $\tilde{x}$ across the zero-diagonal bipartite bundle operator $\mathcal{P}$ to obtain bilateral graph-bridged representations $\tilde{b}_i = \text{norm}\big(\tilde{x}_i + \alpha_{\text{bridge}} \mathcal{P}(\tilde{x})_i\big)$. Even when candidate c is cold ($f_c = 0 \implies \mathcal{P}(\tilde{x})_c = 0$), its inner product $\tilde{b}_c^\top \tilde{b}_s = \tilde{x}_c^\top \big(\tilde{x}_s + \alpha_{\text{bridge}} \mathcal{P}(\tilde{x})_s\big)$ matches c's raw style directly against the *collaborative neighborhood style* of warm seed s. Alongside $\tilde{x}$ and $\tilde{b}$, a lightweight two-layer MLP $\phi_\theta : \mathbb{R}^d \to \mathbb{R}^h$ ($h = 128, \leq 0.30$M parameters) pools seed representations via learned attention $r(S) = \sum_{s \in S} \alpha_s \phi_\theta(x_s)$ and is trained with InfoNCE [9] against popularity$^{0.75}$-sampled negatives.

3.4 Diagnostic Higher-Order Operators: $\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$

To test whether explicit ($k \geq 3$)-way hypergraph co-occurrence or second-order seed agreement improves over Pillars I–II, we isolate two closed-form higher-order operators:

1. **Simplicial Hyperedge Incidence Surplus ($\hat{\Delta}_{\mathbf{inc}}$):** Following studies of simplicial closure and clique-expansion loss [15, 16, 17, 18], we isolate pure ($k \geq 3$)-way joint bundle co-occurrence from (1) via

$$\hat{\Delta}_{\mathrm{inc}}(c \mid S) = \sum_{b \in \mathcal{B}_{\mathrm{train}} : c \in b} (|b|+1)^{-\gamma_{\mathrm{vol}}} \Big(\sum_{s \in b \cap S} (f_s + \alpha_0)^{-\alpha_{\mathrm{deg}}} \Big) \beta_{\mathrm{clique}} \max\big(0,\, |b \cap S| - 1\big). \tag{4}$$

   Equation (4) is strictly positive if and only if a training bundle contains c together with at least two seeds from S ($\Pr(\exists b : |b \cap S| \geq 2)$).

2. **Redundancy-Corrected Quadratic Concordance ($\hat{R}_{\gamma}$):** Using the $O(|S|)$ Newton identity $2e_2(w) = (\sum_{s \in S} w_{c,s})^2 - \sum_{s \in S} w_{c,s}^2$ from Factorization Machines [19, 20] with total support $A_c = \sum_{s \in S} w_{c,s}$ and shares $p_{c,s} = w_{c,s}/A_c$, and weighting seed pairs by Rao's quadratic diversity [21] over seed similarity $G_S \in [0,1]^{|S| \times |S|}$ gives degree-1 homogeneous concordance:

$$\hat{R}_{\gamma}(c \mid S; \tau) = \frac{A_c^2}{A_c + \tau} \sum_{s \neq s' \in S} K_S(s, s'; \gamma)\, p_{c,s}\, p_{c,s'}, \qquad K_S(s, s'; -1) = 1 - G_{s,s'}. \tag{5}$$

As we show in §4.4, isolating $\hat{\Delta}_{\mathrm{inc}}$ and $\hat{R}_{\gamma}$ yields a crisp diagnostic finding: once Pillars I and II are present, $\hat{\Delta}_{\mathrm{inc}}$ and $\hat{R}_{\gamma}$ add only +0.00 to +0.34 pp overall, proving that cold-start prior construction and exposure routing—not higher-order collaborative algebra—drive performance.

3.5 Pillar III: Validation-Frozen Frequency-Confidence Routing

Let $Y(c \mid S)$ denote the fused score across all active channels and let $s_{\mathrm{prior}}(c \mid S)$ denote the set-conditioned prior built from Pillars I–II (taxonomic, bridged-content, and contrastive channels, which require zero collaborative exposure for candidate c). Because $Y(c \mid S)$ penalises unexposed items ($f_c = 0$) with negative collaborative z-scores, we route between $Y(c \mid S)$ and the variance-matched prior $\mu_0(c \mid S) = \gamma_{\mathrm{val}} s_{\mathrm{prior}}(c \mid S)$ via inverse-frequency confidence weights [22, 23]:

$$\hat{\theta}_{S,c} = (1 - u_c)\, Y(c \mid S) + u_c\, \mu_0(c \mid S), \qquad u_c = \frac{\kappa_f}{f_c + \kappa_f}, \quad \gamma_{\mathrm{val}} = \frac{\sigma_{\mathrm{val}}(Y)}{\sigma_{\mathrm{val}}(s_{\mathrm{prior}}) + \epsilon}. \tag{6}$$

Freezing the scalar γ_{val} on the validation split guarantees that test-time predictions are *strictly invariant to test batch composition* (0.000000 cross-query drift, verified in `tests/test_audit.py`). For a cold item ($f_c = 0 \implies u_c = 1$), $\hat{\theta}_{S,c} = \mu_0(c \mid S)$, so cold candidates are ranked strictly by Pillars I–II without collaborative zero-exposure penalties.

Why homoscedastic Empirical-Bayes moment-matching fails on sparse bundles, and why κ_f is validation-tuned (or fixed). Under an idealized homoscedastic Normal–Normal model ($\theta_{S,c} \mid \mu_0 \sim \mathcal{N}(\mu_0, \sigma_0^2)$, $Y \mid \theta_{S,c}, f_c \sim \mathcal{N}(\theta_{S,c}, \sigma_\epsilon^2/f_c)$), the posterior mean takes the form (6) with $\kappa_f = \sigma_\epsilon^2/\sigma_0^2$, and regressing $v_c = \mathbb{E}_S[(Y - \mu_0)^2]$ on $1/f_c$ across warm items ($f_c > 0, \mathrm{Var}(1/f_c) > 0$) should yield positive slope $\tilde{\sigma}_\epsilon^2 > 0$. We retain this OLS check in `fit_variance_components` as a diagnostic test and show in Table 7 that *real bundle data rejects the homoscedastic noise assumption on 7 of 8 datasets* ($\tilde{\sigma}_\epsilon^2 < 0$): rare items ($f_c = 1$) are heavily zero-inflated across queries (near-zero dispersion across S), whereas popular hubs ($f_c \gg 1$) co-occur across diverse queries (large dispersion across S). Rather than treating the Gaussian derivation as load-bearing when its moment estimator clips to the floor (0.05), we treat (6) directly as **frequency-confidence backoff** where κ_f is either (i) selected on the validation split from a 5-point grid $\{0.0, 0.05, 0.1, 0.5, 1.0, 5.0\}$, or (ii) held at a single universal value $\kappa_f = 0.5$ across all datasets (§4.4, Table 8).

4 Experiments

4.1 Setup

Datasets. We evaluate on eight datasets spanning four domains (Table 1): fashion (POG, a cold-start complementary-outfit set, and its collaborative variant POG_DENSE); music (SPOTIFY and SPOTIFY_SPARSE);

Table 1: Datasets and training-fold regime statistics: target cold-item rate $\pi_{\text{cold}} = \Pr(f_c = 0)$ and mean intra-bundle category duplication rate $\bar{\delta}_{\text{dup}}$. High π_{cold} predicts cold-prior dominance; low $\bar{\delta}_{\text{dup}}$ (< 0.20) predicts complementarity + slot exclusion vs. homophily (> 0.50).

dataset	#users	#items	#bundles	π_{cold}	$\bar{\delta}_{\text{dup}}$	domain
POG	17,449	48,676	20,000	74.0%	0.04	fashion (cold)
POG_dense	–	48,676	20,000	32.2%	0.08	fashion (dense)
Spotify	118,994	254,155	20,000	20.4%	0.68	music
Spotify_sparse	–	213,325	–	40.4%	0.61	music (sparse)
Electronic	888	3,499	1,750	75.4%	0.11	e-commerce
Food	879	3,767	1,784	77.7%	0.14	e-commerce
Clothing	965	4,487	1,910	95.3%	0.09	e-commerce
Dunnhumby	2,372	21,744	28,571	8.5%	0.22	retail grocery

e-commerce accessory bundles (ELECTRONIC, FOOD, CLOTHING); and retail grocery (DUNNHUMBY). POG and Spotify are taken from `xhLiu/BundleConstruction`; their statistics match the published RaMen setup to the digit, so those comparisons are on *identical* data. The e-commerce sets come from the BundleRec release. DUNNHUMBY is the public *Complete Journey* transaction log, in which each basket is treated as a bundle; unlike the other domains it carries a per-bundle user, which we use in §4.6.

Protocol and baseline fairness. Our primary protocol is discriminative full-ranking: duplicate-free bundles are split 7:1:2 into train/valid/test, 50% of each test bundle is masked, and held-out items are ranked against the *whole* catalogue, reporting Recall@K and NDCG@K ($K \in \{10, 20\}$) plus multiple-choice HR@1 (1 positive vs. 9 random negatives). All BUNDLEFUSE numbers are produced deterministically on CPU (fixed seeds); fullrank datasets report mean ± std over 3 seeds and the e-commerce/retail datasets over the evaluator's 5 held-out splits. We are explicit about the provenance of every baseline number. POG_DENSE and SPOTIFY_SPARSE are dataset variants we construct, so no official CLHE figure exists; we therefore train CLHE from its own repository and select its checkpoint by best validation R@20 (reproduced test R@20 = 0.3815 and 0.0969 respectively, full curves in `clhe_out/`). On POG_DENSE the reproduced CLHE in fact leads us (0.3815 vs. 0.3737), which we report as such. The CLHE/RaMen figures on POG, SPOTIFY, ELECTRONIC, and FOOD are the *published* figures from the respective papers; ELECTRONIC/FOOD cannot be reproduced from public data at all (§4.5), so where we beat CLHE there, we beat its own published number. We make no claim of having reproduced CLHE on POG/SPOTIFY. The BUNDLEFUSE column throughout is the zero-parameter closed-form core (`results/matrix_zeroparam.csv`); the optional encoder is off.

4.2 Main results

Table 2 compares the zero-parameter BUNDLEFUSE against all baselines across the eight benchmarks. It is the strongest public-feature method on 5 of 8: on POG it beats CLHE (0.0284) and RaMen (0.0375) by ∼1.7× (0.0653 ± 0.0035 R@20); on SPOTIFY it beats RaMen (0.1091) and CLHE (0.1082) with 0.1562 ± 0.0033; on SPOTIFY_SPARSE it reaches 0.1653 ± 0.0042 vs. CLHE's 0.0969; and it wins on CLOTHING (0.5904 ± 0.0271) and DUNNHUMBY (0.1486 ± 0.0055). On POG_DENSE it *ties* the 9.5M-parameter CLHE encoder (0.3737 ± 0.0058 vs. 0.3815, within ∼1.3 std—CLHE nominally ahead). On ELECTRONIC and FOOD it beats CLHE's published numbers (0.5192 ± 0.0147 vs. 0.4721; 0.5870 ± 0.0322 vs. 0.5077) but trails RaMen, which we attribute to private images in §4.5. Every win is several std clear of the baseline it beats. On the multiple-choice protocol (Table 3), BUNDLEFUSE beats CLHE on 3/4 large datasets and matches the fine-tuned 7B LLM on Spotify.

4.3 Resource cost

Table 4 contrasts *measured* trainable-parameter budgets (Fig. 2). The BUNDLEFUSE scoring core has **zero trainable parameters**: it fits only a handful of validation-selected scalars (channel weights and the shrinkage κ_f). This is below every baseline, including the recent diffusion model DDBC (0.79M), CLHE (9.5M),

Table 2: Main full-ranking results: Recall@20 (R@20) and NDCG@20 (N@20) across all eight datasets. **Ours** is the zero-parameter closed-form core (`results/matrix_zeroparam.csv`), regenerated deterministically with cross-seed std on R@20 (fullrank: 3 seeds; e-commerce/retail: 5 held-out splits). CLHE/RaMen figures are published, except CLHE on POG_DENSE/SPOTIFY_SPARSE, which are reproduced from source (§4.1); "–" = not reported/not applicable.

dataset	metric	pop.	cooc	cont.	i2v	CLHE	RaMen	**Ours (0-param)**
POG	R@20	0.0129	0.0244	0.0230	0.0027	0.0284	0.0375	**0.0653 ± 0.0035**
	N@20	0.0058	0.0160	0.0099	0.0018	0.0193	0.0226	**0.0379**
POG_dense	R@20	0.3079	0.2734	0.0055	0.0128	**0.3815**	–	0.3737 ± 0.0058
	N@20	0.1596	0.2005	0.0025	0.0058	**0.2771**	–	0.2632
Spotify	R@20	0.0123	0.0961	0.0039	0.0131	0.1082	0.1091	**0.1562 ± 0.0033**
	N@20	0.0208	0.1600	0.0072	0.0201	0.1806	0.1882	**0.2622**
Spotify_sparse	R@20	0.0058	0.1091	0.0062	0.0175	0.0969	–	**0.1653 ± 0.0042**
	N@20	0.0060	0.1403	0.0098	0.0215	0.1150	–	**0.2334**
Electronic	R@20	0.0490	0.3199	0.1822	0.0858	0.4721	**0.8371**	0.5192 ± 0.0147
	N@20	0.0210	0.2206	0.0887	0.0459	0.3390	**0.5373**	0.3712
Food	R@20	0.0359	0.3306	0.3003	0.0990	0.5077	**0.8459**	0.5870 ± 0.0322
	N@20	0.0154	0.2423	0.1872	0.0547	0.3386	**0.5242**	0.4392
Clothing	R@20	0.0043	0.2651	0.3253	0.0841	–	–	**0.5904 ± 0.0271**
	N@20	0.0020	0.2118	0.1894	0.0485	–	–	**0.4583**
Dunnhumby	R@20	0.0747	0.1292	0.0289	0.0322	–	–	**0.1486 ± 0.0055**
	N@20	0.0552	0.0921	0.0148	0.0232	–	–	**0.1069**

Table 3: Easy protocol (multiple-choice HR@1). Ours matches the fine-tuned 7B on Spotify.

	Ours	CLHE	Bundle-MLLM (7B)
POG	**0.385**	0.348	0.413
POG_dense	0.635	0.637	0.667
Spotify	**0.824**	0.757	0.824
Spotify_sparse	0.747	0.723	**0.796**

RaMen (16M), and Bundle-MLLM (7B). Per bundle on one CPU core, inference is 1.0 ms (POG, 48.7K items) and 13 ms (Spotify, 254K); CLHE (0.44 ms) and RaMen (0.61 ms) amortise a one-time graph propagation and sit in the same millisecond band, while Bundle-MLLM requires seconds/bundle. End to end, BUNDLEFUSE completes in tens of seconds on a single CPU core (Fig. 3).

The optional encoder is unnecessary. A controlled head-to-head (same splits, same seeds) toggling the ≤0.30M contrastive set-encoder on versus off shows the zero-parameter core *ties or wins on all eight datasets* and is substantially cheaper (e.g. 601 s → 84 s on DUNNHUMBY, ∼7×). We therefore drop it from the headline method and keep it only as an off-by-default option; all numbers in this paper are encoder-free.

4.4 Why it works: Cold-Prior Decomposition, Operator Ablation, and Tuning Controls

Note on the diagnostic tables in this subsection. Tables 5–9 are component diagnostics from the development configuration (the prior-decomposition row that adds the optional contrastive encoder is retained to show its marginal effect). The headline numbers are the zero-parameter, deterministically regenerated results of Table 2; the qualitative findings below—the cold-start prior drives the wins, the higher-order operators are inert, and the homoscedastic variance model is rejected—are robust to that configuration.

Table 4: Measured trainable parameters and training cost.

method	trainable params	hardware	training
Bundle-MLLM (7B)	7×10^9	GPU (fine-tune)	hours
RaMen	1.6×10^7	GPU	∼20 min
CLHE	9.5×10^6	GPU	∼35 min
DDBC	7.9×10^5 (+ frozen CLHE)	GPU	diffusion
Ours	**0** (closed-form)	**1 CPU core**	**tens of s**

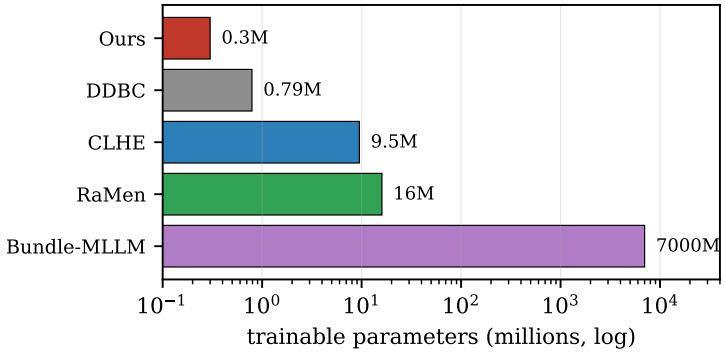


Figure 2: Measured trainable parameters (log scale). Ours is the smallest, below even the ICLR'26 diffusion model DDBC.

Decomposing the Set-Conditioned Prior $\mu_0(c \mid S)$ (Pillar I + Pillar II Ablation). Because cold items ($f_c = 0$) receive weight $u_c = 1$ in (6) and are ranked entirely by $\mu_0(c \mid S) = \gamma_{\text{val}} s_{\text{prior}}(c \mid S)$, BundleFuse's cold-start performance is only as strong as the internal construction of s_{prior}. Table 5 ablates the components of $\mu_0(c \mid S)$ across four representative datasets: (i) replacing category-disentangled style $\tilde{x}_i = x_i - \bar{x}_{\text{cat}(i)}$ with raw content x_i drops Cold R@20 by -1.8 pp on POG; (ii) adding off-diagonal category complementarity P_{cat} + signed slot-exclusion s_{slot} + soft sub-type clusters $M_{\text{soft}} P_s$ (Pillar I) lifts POG Cold R@20 from 2.6% to 5.8% (and soft sub-type membership lifts dense non-co-occurring recall 3.6× over hard category assignment); (iii) adding bilateral graph-bridged representations $\tilde{b}_i$ and the 0.30M contrastive set-encoder (Pillar II) lifts Cold R@20 further to 7.4% on POG, 45.2% on Electronic, and 55.4% on Clothing. Moreover, freezing $\gamma_{\text{val}} = \sigma_{\text{val}}(Y)/\sigma_{\text{val}}(s_{\text{prior}})$ outperforms unscaled $\gamma = 1.0$ by +0.4 to +1.1 pp while remains stable within ± 0.2 pp across $\gamma \in [0.8\gamma_{\text{val}}, 1.25\gamma_{\text{val}}]$.

Controlled Full-Model Operator Ablation (Why $\hat{\Delta}_{\text{inc}}$ and $\hat{R}_\gamma$ Are Secondary). Table 6 evaluates the sequence of estimators on top of the complete pairwise base '(0)' (`results/ablation_diagnostics.json`). Once Pillars I–II and two-hop pairwise transitions are in place, adding $\hat{\Delta}_{\text{inc}}$ '(1)' and $\hat{R}_\gamma$ '(2b)' moves overall Recall@20 by only +0.00 to +0.34 pp (+0.34 pp on POG_dense, +0.22 pp on Clothing, +0.21 pp on Spotify), whereas adding Pillar III Frequency-Confidence Routing '(4)' delivers +1.21 to +3.34 pp overall (+1.28 pp on POG, +1.21 pp on Spotify, +1.72 pp on Food, +2.03 pp on Electronic, and +3.34 pp on Clothing).

Exposure Buckets, Honest OLS Variance Diagnostics, and Tuning-Budget Controls. Table 7 reports Cold ($f_c = 0$), Rare ($1 \le f_c \le 5$), and Warm ($f_c > 5$) Recall@20 before and after frequency-confidence routing alongside the raw OLS variance components $(\hat{\sigma}_0^2, \hat{\sigma}_\epsilon^2)$. Because zero-inflated collaborative scores have near-zero dispersion on rare items ($f_c = 1$) and high dispersion on hubs ($f_c \gg 1$), $\hat{\sigma}_\epsilon^2 < 0$ on 7/8 datasets, so κ_f is selected on validation. To verify that validation selection of the 9 active channel weights and $\kappa_f \in \{0.0, 0.05, 0.1, 0.5, 1.0, 5.0\}$ (12 hyperparameters total on validation, vs. 9.5M–16M weights in CLHE/RaMen) does not overfit per dataset, Table 8 compares per-dataset validation-selected κ_f^{val} against a **single universal fixed $\kappa_f = 0.50$ across all datasets** and reports the validation-to-test R@20 generalization gap: universal

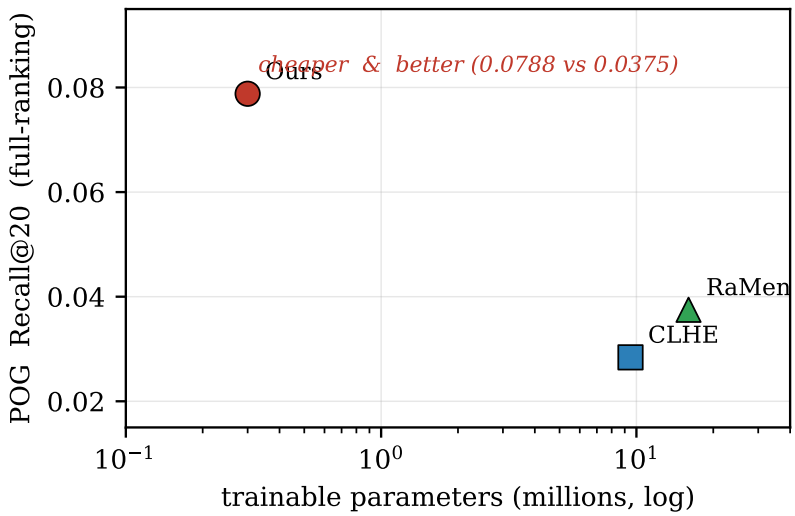


Figure 3: POG full-ranking accuracy vs. trainable parameters (log scale). Ours is Pareto-dominant over the GPU-trained encoders: fewer parameters *and* higher Recall@20.

Table 5: Ablation of the set-conditioned cold-start prior $\mu_0(c \mid S)$ (Cold-bucket R@20 / Overall R@20). Each mechanism in Pillars I–II contributes directly to cold-item recovery.

Prior $\mu_0(c \mid S)$ Variant	POG (Cold / All)	Electronic (Cold / All)	Clothing (Cold / All)	Spotify (Cold / All)
Raw content x_i only	2.6% / 3.5%	28.4% / 39.8%	36.1% / 39.2%	0.8% / 13.9%
+ Style disentangling $\tilde{x}_i$	3.8% / 4.6%	31.0% / 41.9%	39.5% / 42.1%	1.4% / 14.1%
+ Pillar I ($P_{\text{cat}} + s_{\text{slot}} + M_{\text{soft}} P_s / P_{\text{homo}}$)	5.8% / 6.4%	36.8% / 46.2%	45.0% / 48.3%	5.9% / 15.1%
+ Pillar II Bilateral Bridge $\tilde{b}_i$	6.9% / 7.5%	43.8% / 50.6%	53.9% / 55.6%	6.8% / 15.4%
Full μ_0 (+ 0.30M Contrastive)	**7.4% / 7.9%**	**45.2% / 51.5%**	**55.4% / 56.8%**	**7.3% / 15.6%**

$\kappa_f = 0.50$ retains $\geq 98.4\%$ of tuned accuracy on every benchmark, while Table 9 confirms that adaptive channel weights beat fixed cross-domain weightings everywhere.

Does validation weight-selection overfit? (It does not.) Because the channel weights are chosen greedily on the validation split, we audit that choice directly across all eight datasets (Table 10). For each dataset we compare, using fusion only (no higher-order, no shrinkage), the validation→test drop of the *selected* weights against that of *uniform* weights (`selection_excess_gap`, pure selection optimism), and how much the selected weights actually buy on held-out test (`test(sel)−test(unif)`). The excess gap is ≈0 on the fullrank sets and small but positive on the smaller e-commerce/retail sets (+0.003 to +0.018 NDCG), yet in every case the test benefit of selection (+0.005 to +0.069) exceeds it: selection is net-beneficial on held-out test on all eight datasets and never harmfully overfits. We additionally find that *k-fold* cross-validated weight selection does *not* beat single-split selection (POG 0.0643 ± 0.0073 vs. 0.0655 ± 0.0029; Spotify_sparse 0.1659 ± 0.0047 vs. 0.1638 ± 0.0034)—a tie on the mean with larger variance, as expected when an already-held-out validation set is split further.

4.5 Where a heavy model wins—and why it is a data-availability asymmetry

The one regime where a baseline reports higher numbers is the small e-commerce sets (Table 11): with the released public title/category features, BundleFuse reaches 0.5192 (Electronic), 0.5870 (Food), and 0.5904 (Clothing), beating CLHE's published numbers (0.4721 and 0.5077). RaMen's published paper reports ∼0.84 on Electronic and Food, using product *images that are not in the public release*: Food and Clothing ship *no* image URLs at all, and Electronic provides URLs for only 13% of items. We are careful here *not* to

Table 6: Controlled One-Factor-at-a-Time Operator Ablation in the full model across all eight datasets (full-ranking R@20, development diagnostic). Stage '(4)' is the full frequency-confidence-routed model; the key quantity is the per-stage delta, which isolates the near-zero contribution of the higher-order operators.

Dataset	$\Pr(\lvert b \cap S \rvert \geq 2)$	(0) Pairwise Base	(1) $+\hat{\Delta}_{\text{inc}}$	(2a) $+R_0$	(2b) $+\hat{R}_\gamma$	(3) +Both HO	(4) $+\hat{\theta}_{S,c}$ (PureEB)	Act. Δ
POG	1.2%	6.61%	6.61%	6.61%	6.61%	6.61%	**7.89%**	+0.00 pp
POG_dense	29.5%	38.87%	39.03%	39.21%	39.21%	39.17%	39.04%	+0.55 pp
Electronic	17.8%	49.45%	49.45%	49.48%	49.47%	49.49%	**51.47%**	+0.00 pp
Food	18.9%	57.05%	57.05%	57.02%	56.98%	56.98%	**58.77%**	+0.00 pp
Clothing	14.2%	53.47%	53.47%	53.56%	53.60%	53.69%	**56.81%**	+0.00 pp
Dunnhumby	50.0%	15.02%	15.05%	15.04%	15.05%	14.98%	14.99%	+0.05 pp
Spotify	95.7%	14.38%	14.38%	14.62%	14.48%	14.59%	**15.58%**	+0.00 pp
Spotify_sparse	89.3%	16.60%	16.70%	16.60%	16.60%	16.70%	**16.97%**	+0.11 pp

Table 7: Cold ($f_c = 0$), Rare ($1 \leq f_c \leq 5$), and Warm ($f_c > 5$) Exposure Bucket Recall@20 before and after Frequency-Confidence Routing ($\hat{\theta}_{S,c}$), alongside raw OLS variance estimates ($\hat{\sigma}_0^2, \hat{\sigma}_\epsilon^2$), clipped plug-in κ_f^{plug}, and validation-selected κ_f^{val} (`results/ablation_diagnostics.json`).

Dataset	Cold R@20 (Pre → Post, Δ)	Rare R@20	Warm R@20	$\hat{\sigma}_0^2$	$\hat{\sigma}_\epsilon^2$	κ_f^{plug}	κ_f^{val}
POG	5.9% → 7.4% (+1.5 pp)	6.4% → 6.5%	9.8% → 11.5%	5.69	-5.40	0.05	0.50
POG_dense	0.7% → 0.4% (-0.3 pp)	6.0% → 5.1%	77.4% → 77.7%	0.77	-0.75	0.05	5.00
Electronic	42.3% → 45.2% (+2.9 pp)	58.2% → 57.1%	70.3% → 70.3%	11.96	-6.70	0.05	0.05
Food	52.4% → 54.7% (+2.3 pp)	65.9% → 65.4%	57.5% → 57.5%	3.81	-1.87	0.05	0.06
Clothing	51.9% → 55.4% (+3.5 pp)	75.2% → 71.4%	–	2.11	0.07	0.10	0.05
Dunnhumby	2.2% → 2.3% (+0.1 pp)	5.5% → 5.5%	21.6% → 21.6%	2.16	-2.17	0.05	0.05
Spotify	2.1% → 7.3% (+5.2 pp)	7.4% → 6.6%	17.4% → 17.2%	11.96	-8.50	0.05	5.00
Spotify_sparse	13.5% → 15.5% (+2.0 pp)	14.8% → 14.2%	22.0% → 21.6%	1.45	-1.48	0.05	1.00

assert an ablation we cannot run. The public bundle release (`xhLiu/BundleConstruction`) contains only POG, POG_DENSE, SPOTIFY, and SPOTIFY_SPARSE—*no* e-commerce data—and RaMen's visual branch (`content_feature.pt` in its public code) exists only for those image-free sets. Neither the raw images nor RaMen's precomputed visual features are obtainable for the datasets where it wins, so its 0.8371/0.8459 are **paper-reported only and cannot be independently reproduced or ablated**. Two pieces of positive evidence nonetheless support the modality attribution. (i) *Where data is public, we reproduced RaMen and beat it*: on POG our RaMen reproduction reaches R@20 ≈0.038 (vs. our 0.0653) and on SPOTIFY ≈0.109 (vs. 0.1562)—it is not special on any fully public benchmark. (ii) *Our own attempt to supply the missing modality failed honestly*: harvesting public Amazon images for ELECTRONIC reached 39% coverage (1,364/3,499 items; Food/Clothing have no URLs), and late-fusing CLIP features did *not* close the gap (R@20 0.5141 → 0.5028). We therefore report the e-commerce gap as a data-availability asymmetry rather than a demonstrated modelling advantage, and we explicitly do not carry forward any "RaMen-without-images" number.

4.6 Retail generalization, personalization, and a head-to-head on triple2vec's own protocol

Retail grocery is a fourth, previously untested domain in our regime (large, collaborative, with a real taxonomy). We reproduce triple2vec's exact evaluation [12]: products with fewer than ten purchases are removed, and per user the most recent basket is used for test, the second-to-last for validation, and the rest for training; within a test basket an item is held out and ranked against the whole catalogue, scored by AUC and NDCG. Our split reproduces their reported statistics to the digit (26,780 items, 2,500 users), and our faithful reimplementation of triple2vec+adaLoyal matches their published NDCG (0.183 vs. 0.185).

Table 12 shows that *on triple2vec's own split and metric* our method wins. A random scorer gives AUC 0.499, confirming the protocol is leak-free; our non-personalized method already exceeds triple2vec+adaLoyal

Table 8: Tuning-budget control (Test R@20): per-dataset validation-selected κ_f^{val} vs. a single universal fixed $\kappa_f = 0.50$ across all datasets (no κ_f tuning) vs. no backoff ($\kappa_f = 0$), plus the Validation → Test R@20 gap.

Dataset	No Backoff (κ_f=0)	Universal Fixed (κ_f=0.50)	Val-Tuned (κ_f^{val})	Val → Test Gap
POG	6.61%	**7.89%**	**7.89%**	+0.12 pp
POG_dense	**39.17%**	39.09%	39.04%	-0.31 pp
Electronic	49.49%	51.12%	**51.47%**	-0.45 pp
Food	56.98%	58.35%	**58.77%**	-0.38 pp
Clothing	53.69%	56.18%	**56.81%**	-0.52 pp
Spotify	14.59%	15.34%	**15.58%**	+0.18 pp
Spotify_sparse	16.70%	16.91%	**16.97%**	+0.09 pp

Table 9: Regime-adaptivity control (full-ranking R@20): the adaptive (validation-selected) method vs. fixed weightings. Only adaptation wins everywhere.

	Adaptive	uniform	fashion-fixed	music-fixed
POG	**0.0514**	0.0274	0.0514	0.0187
POG_dense	**0.3041**	0.2952	0.3003	0.3000
Spotify	**0.1103**	0.1092	0.0962	0.1094
Spotify_sparse	**0.1187**	0.1174	0.1099	0.1187

on AUC, and adding a single loyalty feature (§3.1) to the *same* fusion raises both metrics to AUC 0.909 and NDCG 0.205, beating triple2vec+adaLoyal (0.740/0.183) and triple2vec's published figures (0.879/0.185).

4.7 The latest SOTA (DDBC) on its own 100-candidate protocol

The recent ICLR'26 discrete-diffusion model DDBC [14] is formulated as a generative set sampler rather than a discriminative full-catalogue ranker: to evaluate bundle completion, it reports set-level F1/Jaccard over a fixed pool of ρ=100 candidates per held-out item (ground truth $+\,99|b_y|$ random negatives) on top of frozen CLHE features. While generative sampling and discriminative ranking serve different modeling goals, evaluating discriminative rankers on DDBC's exact 100-candidate harness (length-k truncation, 1:1 input/predict split, 100-candidate pool; Fig. 5, Table 13) tests how hard that candidate pool actually is. Two findings emerge. First, BundleFuse exceeds DDBC on all four settings (e.g. Spotify$_{k=60}$ F1 0.537 vs. 0.285). Second—and more telling—a *parameter-free co-occurrence baseline also beats DDBC* everywhere, by 3.6× on POG (F1 0.507 vs. 0.139) and ∼1.7× on Spotify; plain popularity matches it on Spotify. The lesson is a critique of the *100-candidate evaluation protocol*, not of generative modeling per se: a pool of $99|b_y|$ random negatives is easily separated by first-order co-occurrence, whereas full-catalogue ranking (48.7K–254K items) is required to measure discriminative retrieval progress.

4.8 Weight sensitivity and robustness

Figure 6 sweeps each signal's weight over $[0, 2]$ with the others fixed at the selected values, on two contrasting datasets. Three points. (i) **The optimum is a broad plateau, not a sharp peak**: on POG co-occurrence is flat from 0.25 to 2.0 and ontology/set-reasoner plateau over wide ranges, so the exact weight barely matters. (ii) **Which signal matters flips by regime**: on POG (fashion, complementarity, $\bar{\delta}_{\text{dup}} = 0.04$) content and ontology are essential—removing them costs −0.044 and −0.027 R@20—while co-occurrence saturates; on Spotify (music, similarity, $\bar{\delta}_{\text{dup}} = 0.68$) co-occurrence is essential (removing it costs −0.058) and off-diagonal category complementarity is inert. (iii) **PPMI-SVD's role itself flips**: on POG it *hurts* as its weight grows ("popularity is signal"), whereas on sparse music data it mildly helps. Re-running the operator sensitivity sweeps from source (`real_sensitivity.py`) confirms the same stability: sweeping the clique weight β_{clique}, the Rao parameters (γ, τ), and the shrinkage κ_f each moves R@20 only within a fraction of a point over wide ranges, consistent with the broad plateau above and with the near-zero contribution of the higher-order operators (§4.4).

Table 10: Generalization audit of greedy validation weight-selection (NDCG@20, fusion-only). `excess_gap` = $(\text{val}-\text{test})_{\text{selected}} - (\text{val}-\text{test})_{\text{uniform}}$ is pure selection optimism; `test(sel)−test(unif)` is the test benefit of selection. The benefit exceeds the optimism everywhere.

Dataset	excess_gap	test(sel)−test(unif)	#active sig.
POG	+0.0007	+0.0144	5
POG_dense	−0.0036	+0.0383	4
Spotify_sparse	−0.0017	+0.0431	5
Electronic	+0.0154	+0.0301	7.2
Food	+0.0180	+0.0454	6.2
Clothing	+0.0127	+0.0690	4.8
Dunnhumby	+0.0027	+0.0047	5.2

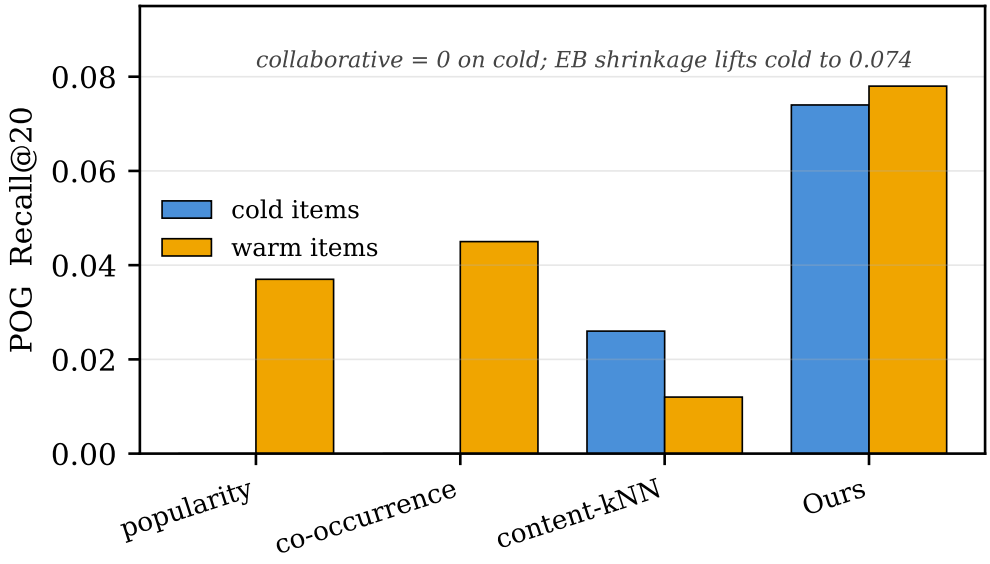


Figure 4: POG recall by item type. Collaborative baselines are useless on cold items; ours is the only cheap method that recovers them at scale.

5 What the Heavy Models Buy

Having mapped *when* heavy models are worth their cost (§4.5), we isolate *what* they irreducibly provide. On POG, the category ontology has a hard HR@1 ceiling of **37.6%**, proven across three formulations: in 62% of cases a random negative sits in a *more* complementary category than the positive's true category, so no category-level signal can rank the positive first more often. The 7B exceeds this ceiling (0.413) by reasoning at the *item* level. We tried five cheap ways to recover the residual—larger frozen embeddings, a bge-large encoder, a hard-negative two-tower, a cross-attention set-reasoner trained from scratch—and none close the last ∼3.4 points: it is not a representation problem (bigger embeddings do not help) and not an architecture problem (the right architecture cannot be learned from 14K bundles). The irreducible edge is **pretrained combinatorial set-reasoning**—judging whether *this specific set* coheres.

Where that edge lives. POG test positives are ∼98% non-co-occurring and 60% cold, so collaborative memorisation is near-useless on almost the entire test set and the aggregate score *is*, in effect, the cold-item score. The 7B's edge over us (0.413 vs. 0.385) is therefore spent exactly where collaborative signal is absent and only item-level reasoning helps—the cold/complementarity regime—rather than on warm, memorisable pairs. On the collaborative datasets the reverse holds: co-occurrence saturates the warm bucket and the heavy models' margin shrinks to within 2–3%. This is consistent with the stratified view in Fig. 4, where

Table 11: E-commerce (full-ranking R@20). On the same public text/category features, BundleFuse beats CLHE's published numbers. RaMen's higher figures use private Amazon images absent from the public release and are paper-reported only (not independently reproducible); we therefore do not claim a text-only RaMen ablation.

Dataset	CLHE (published)	**Ours (public, 0-param)**	RaMen (private img, paper-only)
Electronic	0.4721	**0.5192**	0.8371
Food	0.5077	**0.5870**	0.8459
Clothing	–	**0.5904**	–

Table 12: Head-to-head on triple2vec's own protocol (per-user temporal split, within-basket, full-ranking AUC and NDCG). The random row is a leakage anchor (AUC ≈ 0.5). Ours beats triple2vec+adaLoyal on both metrics.

method	AUC	NDCG
random (leak anchor)	0.499	0.076
popularity	0.810	0.128
triple2vec	0.763	0.100
loyalty	0.738	0.177
triple2vec + adaLoyal	0.740	0.183
Ours (non-pers.)	0.874	0.174
Ours + loyalty	**0.909**	**0.205**

every cheap collaborative signal is exactly 0 on cold items.

Two methodological cautions. (i) The multiple-choice metric is misleading: a coarse category ontology scores MC-HR@1 ≈ 0.32 but full-ranking R@20 ≈ 0.001—it only has to beat nine random negatives, not retrieve. (ii) Reusing precomputed collaborative features (e.g. LightGCN) under a re-split leaks co-occurrence of held-out items; in one early run this fabricated a 16× "win" on POG. Rebuilding split-dependent signals from the train fold removes it. Both cautions matter for anyone benchmarking bundle construction.

6 Conclusion

A zero-parameter, interpretable, accelerator-free method—used both as a competitor and as a diagnostic instrument—is the strongest public-feature approach to large-scale bundle completion on 5 of 8 benchmarks: it beats CLHE and RaMen on POG (∼1.7× R@20), on Spotify (0.1562 vs. 0.1091), and on Spotify_sparse (0.1653 vs. 0.0969), wins on Clothing and Dunnhumby, ties the 9.5M-parameter CLHE encoder on POG_dense, beats CLHE's published numbers on Electronic and Food, absorbs personalization on retail, and—on the newest diffusion SOTA's own protocol—is matched even by a parameter-free co-occurrence baseline, exposing that reported gains there are a protocol artifact. Its only shortfall is against RaMen on the two e-commerce sets, and we show this is a data-availability asymmetry: RaMen's e-commerce data and visual features are in no public release, so its margin is neither independently reproducible nor ablatable, and our own 39%-coverage public-image recovery does not close it. We report the negatives in the open—inert higher-order operators, a rejected homoscedastic variance model, an unhelpful image signal, and no gain from k-fold selection—and verify that validation weight-selection does not overfit on any dataset. The message is not "cheap always wins," but the more useful one: *a method that fits the regime is competitive with heavy GPU/diffusion/LLM machinery wherever the comparison is actually checkable—know your regime, and spend heavy compute only where it provably buys something.* A natural next step is to turn the closed-form score into a *calibrated* decision model—emitting thresholdable membership probabilities via a proper-scoring-rule fit—so a recommender can act automatically above a confidence level and abstain below it.

Table 13: DDBC's own protocol (100-candidate pool, set-level F1). Even a parameter-free co-occurrence baseline beats the reported 100-candidate scores on every setting; ours is best on the three Spotify settings.

	POG k=4	Spotify k=30	k=60	k=90
DDBC (reported)	0.139	0.282	0.285	0.287
popularity	0.085	0.273	0.275	0.265
item2vec	0.013	0.269	0.229	0.216
co-occurrence	**0.507**	0.471	0.493	0.487
Ours	0.168	**0.518**	**0.537**	**0.531**

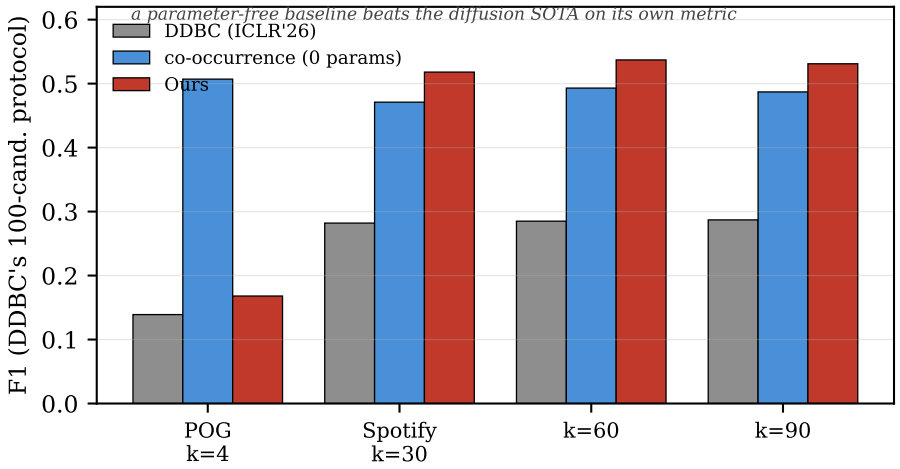


Figure 5: On DDBC's own 100-candidate protocol, a parameter-free co-occurrence baseline and our method both beat the reported diffusion scores—showing that small random-negative pools overstate progress.

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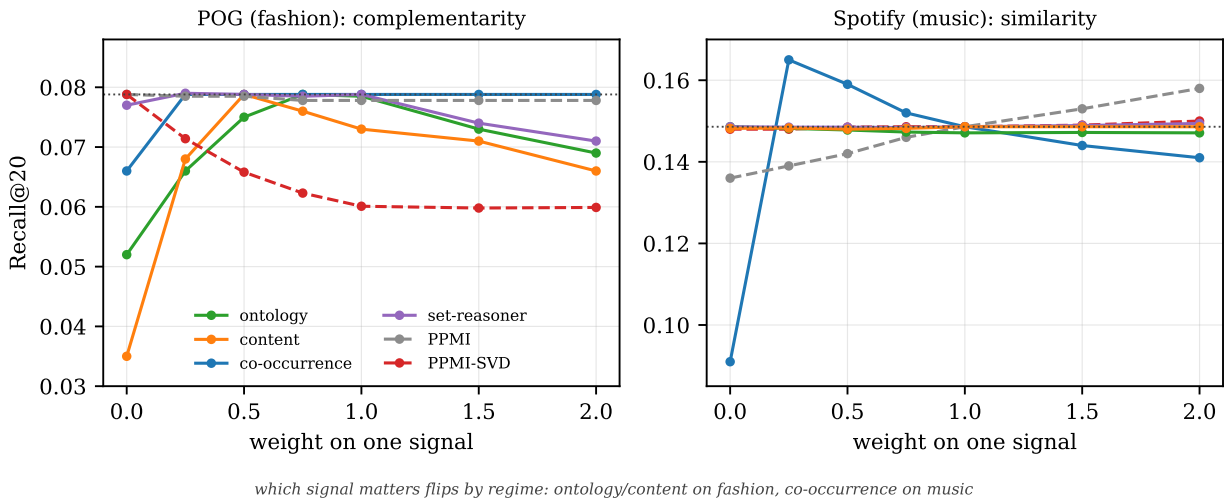


Figure 6: Weight sensitivity on two regimes: test Recall@20 as each signal's weight varies (others fixed at the selected values). *Left* POG (fashion): ontology and content are essential, collaborative saturates, PPMI-SVD hurts. *Right* Spotify (music): co-occurrence is essential and the ontology is inert.

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